

The Erosion of SaaS Moats via AI Coding Agents

Vedang Ratan Vatsa

vedangvats@gmail.com

ABSTRACT

For over two decades, the primary barrier to entry in the Business-to-Business software market has been the high marginal cost of software engineering. This paper presents a structural economic analysis of how autonomous AI coding agents are systematically dismantling this barrier. Drawing on empirical productivity data from GitHub Copilot (Peng et al., 2023) and the SWE-bench evaluation framework for autonomous software engineering (Jimenez et al., 2023), the analysis demonstrates that the cost of generating production-grade boilerplate code is trending toward zero. This rapid deflation in engineering costs fundamentally alters the enterprise “build versus buy” calculus. The paper examines leading indicators of this shift, notably the public decision by enterprise organizations such as Klarna to deprecate tier-one Software-as-a-Service subscriptions in favor of bespoke, AI-generated internal tooling. The analysis concludes that as code generation becomes fully commoditized, traditional Software-as-a-Service valuations will face severe compression. Competitive moats will subsequently shift away from software execution and concentrate entirely on proprietary datasets, systemic distribution channels, and entrenched workflow lock-in.

Keywords: AI coding agents, SaaS economics, software engineering, competitive moats, build versus buy, SWE-bench

1. INTRODUCTION

The economic foundation of the Software-as-a-Service (SaaS) industry rests on a simple premise. Building and maintaining enterprise software is highly complex and prohibitively expensive for most non-technology organizations. Consequently, businesses willingly pay recurring subscription fees to outsource this complexity to specialized vendors. The vendor amortizes the high fixed cost of software engineering across thousands of customers, creating a high-margin, scalable business model protected by a deep technical moat.

This economic equilibrium is currently being disrupted by the rapid advancement of large language models and autonomous coding agents. Tools designed to generate,

debug, and deploy code are structurally reducing the marginal cost of software engineering.

This paper examines the economic consequences of this deflationary pressure. It begins by analyzing empirical data on developer productivity gains. It then explores how these gains alter the enterprise purchasing calculus, leading to a resurgence of internal software development. The analysis concludes by projecting the future of competitive advantage in a post-code digital economy.

2. THE COLLAPSE OF THE SOFTWARE ENGINEERING BOTTLENECK

The velocity of software creation has historically been bottlenecked by human cognitive capacity and typing speed. AI augmentation has demonstrably removed these constraints.

In a controlled experiment conducted by Microsoft Research and MIT, developers utilizing GitHub Copilot completed a standard HTTP server construction task 55% faster than a control group operating without AI assistance (Peng et al., 2023). While this study measured human-in-the-loop augmentation, subsequent research points toward full autonomy.

The SWE-bench framework, developed by researchers at Princeton University, evaluates the ability of language models to autonomously resolve real-world GitHub issues (Jimenez et al., 2023). The framework measures whether an agent can navigate a codebase, identify a bug, write the necessary patch, and pass all associated unit tests without human intervention. The rapid improvement in SWE-bench resolution rates across successive model generations indicates that autonomous software maintenance is becoming a viable enterprise capability.

When an AI agent can autonomously resolve a Jira ticket for cents on the dollar compared to a human engineer, the fundamental cost structure of software development collapses. The technical moat that previously protected incumbent SaaS vendors effectively evaporates.

3. THE ECONOMIC REVERSAL OF BUILD VERSUS BUY

The deflation of engineering costs directly impacts enterprise procurement strategy. For decades, the consensus strategy for enterprise IT was to “buy” rather than “build” non-core software systems. The high cost and failure rate of internal software projects made bespoke development irrational.

Autonomous coding agents alter this mathematics. When the cost to build drops dramatically, the friction of vendor lock-in, rigid user interfaces, and compounding subscription fees becomes less acceptable.

This reversal is already materializing in the market. In 2024, the financial technology firm Klarna publicly signaled an aggressive shift away from tier-one SaaS vendors (Klarna, 2024). The company initiated a strategy to deprecate expensive enterprise subscriptions, including systems provided by Salesforce and Workday, in favor of bespoke internal tools generated and maintained by AI.

If a financial services company can replace specialized enterprise software with internally generated code, the broader SaaS market faces an existential threat. The willingness to pay a premium for generic workflow software will compress as the cost of generating custom software approaches zero.

4. THE NEW MOATS

As code generation becomes commoditized, software itself ceases to be a defensible competitive advantage. The value in the technology stack will migrate to layers that cannot be easily replicated by an LLM.

The first durable moat is proprietary data. An AI agent can replicate the user interface of an enterprise CRM in seconds, but it cannot replicate the ten years of proprietary customer interaction data housed within that CRM. The data gravity of incumbent platforms becomes their primary defense against AI-generated competitors.

The second moat is distribution and workflow lock-in. Platforms like Microsoft Office or Slack maintain their position not because their codebase is irreplicable, but because they are deeply embedded in the daily habits and compliance frameworks of millions of workers. Replacing these systems requires overcoming massive organizational inertia, a barrier that cheap code does not address.

Consequently, SaaS vendors that rely purely on functional utility will face extreme pricing pressure and churn. Vendors that leverage network effects, data exclusivity, and regulatory compliance will survive, albeit in a highly altered competitive landscape.

5. CONCLUSION

The advent of autonomous AI coding agents represents a structural shock to the economics of the software industry. By driving the marginal cost of software engineering toward zero, these tools dismantle the technical barriers to entry that have historically protected SaaS incumbents. Empirical data from SWE-bench and developer productivity studies confirm this trajectory. The resulting shift in the “build versus buy” calculus will compress software valuations and force a strategic realignment. In the forthcoming era of commoditized code, competitive advantage will belong exclusively to organizations that control proprietary data and systemic distribution channels.

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